FLEET TEAM DIGEST

737MAX-FTD-32-17021

Issue Title : Main Gear Brake Problems Traced to Overheated Brakes

Airplane Model	ATA	Minor Model(s)
737MAX	3241-41	-7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
737NG	9E991

Originated Date	Last Revised Date	Created On
05/11/2017	05/11/2017	04/21/2017

Estimated Completion Date	Next Update Date

Applicability

All 737MAX airplanes.

Description

A small number of 737NG operators have reported various types of problems with their main gear brakes. Investigation of these problems by Boeing and the brake manufacturer(s) has revealed that many of the problems are due to operational practices that cause overheated brakes. This article discusses the nature of these problems for carbon brakes for all manufacturers. It recommends actions that operators can take to prevent excessive brake temperatures which are consistent with the 737MAX FCTM. This article is not intended to advise that operators deviate from normal or accepted operating procedures.

Background

Boeing has been receiving a small but increasing number of reports from 737NG operators of brake and wheel problems that have been traced to overheated brakes. These problems have affected carbon brakes. Details of typical problems are described below:

A small number of 737NG operators with carbon brakes have experienced brake damage related to hot brakes. The most common problem is called 'thermal oxidation' where the carbon material becomes soft, eventually resulting in structural failure of the carbon disks. Thermal oxidation of carbon occurs after repeated exposure to unusually high temperatures. The photo titled 'thermal oxidation' shows typical oxidation damage in the early stages. If left on-wing, the carbon disks will eventually suffer a structural failure resulting in more severe damage.

Thermal damage can occur to other parts of carbon brakes if overheated, such as the rotor clips, wheel torque bars, adjuster pins, and torque tubes.

Boeing and the brake suppliers have investigated many of these events. In a few cases, Boeing measured actual in-service brake and wheel temperatures on customers' airplanes. Using these measured temperatures, and other information about the flights, Boeing was able to construct a brake thermal profile for the airplane as it flew the day's routes. It was found that relatively minor adjustments to operational practices could provide relief from overheated brakes.

Status

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Closed.

Interim Action

See operator action below.

Final Action

See operator action below.

Operator Action

Boeing suggest the following to prevent high temperatures from occurring:

- We suggest operators reference the 737MAX Flight Crew Training Manual (FCTM) Section 6 which contains numerous suggestions to help minimize brake temperatures.

- We suggest using a lower autobrake setting if appropriate. The affected operators were routinely using higher autobrake settings than were necessary for the conditions. For a typical landing, autobrake setting #3 results in about 50% more kinetic energy for the brakes to absorb than setting #2.

- We suggest using normal levels of reverse thrust on all landings. The affected operators were performing some landings with either no reverse thrust or with idle reverse thrust. For a typical landing, not using reverse thrust results in about twice as much kinetic energy in the brakes than if a normal level of reverse thrust is used.

- The combined effect of high autobrake settings and no reverse thrust can quickly cause thermal damage to the brakes, especially if performed repetitively.

- Residual heat from prior landings is additive to the heat generated from the most recent stop, thereby aggravating the issue.

Related Categories

- Airplane Maintenance Issues : Issues that occur during airplane maintenance or have an impact on airplane reliability
- Best Practices / Operational Efficiency : Operational / Maintenance-focused best practices, techniques, health checks or recommendations for preserving aircraft efficiency.
- In Service Occurrences : null

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